Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Intent-aware Agentic AI for integrated control of multiple unmanned vehicle systems in indoor emergency response with unreliable communication**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical·Materials		Information & Communication	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	Robot Control and Intelligence Technologies	Software	Software Solution
Concept of Proposed Technology	▪ **Core concept** - AI technology for integrated control of multiple unmanned vehicles, such as drones and AMRs, to understand the incident commander's intent and respond effectively to accidents occurring in indoor environments with unstable communication ▪ **Main function** - An LLM-based AI agent that understands the intent of the incident commander for incident response operations - A cooperative game theory-based AI agent for planning effective deployment and operation of multiple unmanned vehicle systems to rapidly grasp indoor spatial configurations			

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	- An AI agent based on preference learning for determining effective communication architectures that minimize communication disruptions through real-time analysis of identified spatial structures ▪ **Expected uses of technology** - Rapidly responding to indoor fire incidents in environments such as buildings or vessels by analyzing structural layouts, ignition causes, and fire origins, and providing critical information to firefighters to support effective fire suppression and minimize casualties - Coordinating multiple unmanned vehicle systems, such as drones and AMRs, to collaboratively explore, monitor, and assess indoor fire scenes - Supporting incident commanders by providing integrated situational awareness and AI-assisted decision-making in communication-constrained indoor environments - Establishing adaptive communication structures among unmanned platforms to maintain situational awareness and command coordination in indoor environments with unreliable communication
Rationale for Joint R&D Project	▪ Advanced expertise in cooperative game theory–based optimization and preference learning is primarily available at leading foreign universities, providing essential theoretical and methodological foundations that are difficult to secure solely through domestic research efforts ▪ In the Republic of Korea, universities and research institutes possess strong capabilities in AI learning architecture design, training execution, and AI algorithm development, enabling complementary and collaborative research in the AI technology development aspects of this project ▪ While many domestic companies possess core technologies for unmanned vehicle system control, their capabilities in AI-based cooperative control of multiple unmanned systems remain limited, necessitating collaborative R&D to advance AI-driven multi-agent control technologies ▪ The development of intent-aware, multi-agent AI systems for indoor emergency response requires the integration of advanced theoretical research and real-world system implementation, which cannot be effectively achieved by a single institution, making international joint R&D essential ▪ Through international joint R&D, this project enables the transfer and localization of advanced AI methodologies into domestic industrial environments, accelerating technology maturation, system validation, and the global competitiveness of Korean SMEs
Key Words	1. Muti-agent AI system

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	2. Intent-aware Agentic AI 3. Preference learning 4. Minimization of communication disruptions 5. Trajectory planning for unmanned vehicle swarms

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	▪ The objective of this joint R&D project is to develop an intent-aware, agentic AI framework for integrated control of multiple unmanned mobile systems to support rapid and effective indoor emergency response under unreliable communication environments. To achieve this goal, the specific R&D objectives are defined as follows. - To develop advanced optimization and decision-making algorithms for multi-agent coordination based on cooperative game theory and preference learning, enabling autonomous and intent-aware behavior of multiple unmanned vehicle systems in complex indoor environments - To design an intent-aware AI agent capable of understanding and interpreting natural-language instructions and operational intent from incident commanders, and translating them into coordinated actions of multiple unmanned platforms during indoor emergency response - To develop AI-based planning and control mechanisms for cooperative operation of heterogeneous unmanned mobile systems, such as drones and autonomous mobile robots (AMRs), enabling efficient exploration, monitoring, and assessment of indoor emergency scenes - To establish adaptive communication-aware control strategies that dynamically determine communication structures and coordination policies among unmanned systems, minimizing performance degradation under unreliable or constrained indoor communication conditions - To validate the proposed AI technologies through simulation-based testing and real-world demonstration scenarios, focusing on indoor emergency response environments such as buildings or vessels - To establish an industry–academia–research collaboration framework that integrates advanced AI methodologies from foreign research institutions with domestic system development, platform integration, and field validation capabilities - To promote technology transfer and localization of advanced AI algorithms and system architectures, enabling practical deployment and commercialization by Korean SMEs in			

	the fields of unmanned systems and AI-based emergency response
R&D Target Specifications	▪ The AI system should accurately interpret natural-language commands and operational intent provided by incident commanders, achieving reliable intent recognition and task translation into coordinated multi-agent actions in indoor emergency scenarios ▪ The system should support coordinated operation of multiple heterogeneous unmanned mobile platforms, including aerial drones and autonomous mobile robots (AMRs), enabling collaborative exploration, monitoring, and assessment of indoor environments ▪ The AI control framework should maintain stable coordination and situational awareness under unreliable or constrained indoor communication conditions by dynamically adapting communication structures and control policies among unmanned systems ▪ The system should provide coarse-to-intermediate level understanding of indoor structural layouts and incident-related information sufficient to support effective emergency response and decision-making ▪ The developed technologies should be integrated into a unified AI control system and validated through simulation-based testing and demonstration in representative indoor emergency response environments, such as buildings or vessels
R&D Plan	▪ **Phase 1: Core AI development and system design (Years 1-2)** - Design and development of core AI algorithms for intent-aware decision-making, including cooperative game theory–based optimization and preference learning for multi-agent coordination - Development of an intent-aware AI agent capable of interpreting natural-language commands and operational intent from incident commanders and translating them into executable multi-agent task plans - Simulation-based validation of core AI algorithms and coordination strategies using representative indoor emergency scenarios ▪ **Phase 2: Integrated System Development and Intermediate Validation (Years 3-4)** - Design of multi-agent coordination and planning mechanisms for heterogeneous unmanned mobile platforms by focusing on cooperative exploration and information gathering - Investigation of communication-aware control strategies to enable robust multi-agent coordination under unreliable or constrained indoor communication conditions - Integration of developed AI algorithms into a unified agentic AI control framework, enabling coordinated operation of multiple unmanned mobile systems - Implementation of adaptive communication structure determination mechanisms to dynamically adjust communication topology and coordination policies in response to environmental and network conditions - System-level performance evaluation focusing on real-time responsiveness and robustness under communication constraints

	- Refinement of AI models and control strategies based on experimental results and feedback from intermediate validation ▪ **Phase 3: Demonstration, Validation, and Technology Transfer (Year 5)** - Demonstration of the integrated intent-aware, multi-agent AI control system in representative indoor emergency response environments, such as buildings or vessel-like structures - Comprehensive validation of system effectiveness, including coordinated unmanned operation, intent interpretation accuracy, communication robustness, and decision-making performance. - Technology transfer and localization activities, enabling domestic SMEs to adopt, customize, and commercialize the developed AI technologies and integrated platforms - Assessment of technology readiness and commercialization potential, including identification of follow-up R&D and deployment opportunities

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	▪ Lead the development of advanced AI decision-making and optimization methodologies, including cooperative game theory–based optimization and preference learning, to support intent-aware and multi-agent coordination ▪ Lead the design of AI system architectures and learning frameworks, including AI model structure design, training pipeline development, and algorithm implementation ▪ Design and validate core algorithms for intent understanding and agentic decision-making, focusing on the interpretation of natural-language commands and high-level operational intent in indoor emergency response scenarios ▪ Provide theoretical guidance and methodological support for multi-agent planning, coordination, and communication-aware control strategies under constrained and unreliable communication environments ▪ Contribute to simulation-based validation frameworks and evaluation methodologies for indoor emergency response scenarios
Expected R&R of Korean SME Partner	▪ Adapt and extend advanced AI methodologies provided by the foreign partner to domestic research and application contexts ▪ Conduct algorithm-level validation and performance analysis through simulation and experimental studies ▪ Support joint research activities, including technical documentation and collaborative publications ▪ Develop and integrate unmanned mobile system control technologies, including platform-level control of autonomous mobile systems ▪ Implement system integration and software–hardware interfacing, enabling seamless operation of AI algorithms on physical unmanned platforms.

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	▪ Conduct field testing, demonstration, and validation studies in representative indoor environments ▪ Lead technology localization, productization, and commercialization efforts, including system packaging, deployment, and maintenance planning.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	▪ **Scientific and Technological Effects** - Advancement of intent-aware and agentic AI technologies by integrating cooperative game theory, preference learning, and multi-agent decision-making for indoor emergency response - Establishment of a robust multi-agent AI control framework capable of coordinating heterogeneous unmanned mobile systems under unreliable and constrained communication environments - Development of scalable and reusable AI architectures applicable to safety-critical multi-agent systems ▪ **Economic and Industrial Effects** - Strengthening the technological competitiveness of Korean SMEs through localization and commercialization of AI-based multi-unmanned-system control technologies - Creation of new industrial opportunities in AI-driven emergency response, safety management, and intelligent unmanned system services - Reduction of economic losses caused by indoor emergencies by improving response efficiency and minimizing damage through AI-assisted decision support ▪ **Policy and Social Effects** - Enhancement of national disaster response capabilities by supporting data-driven and AI-assisted decision-making in indoor emergency situations - Improvement of public safety and protection of human life by reducing exposure to hazardous environments through unmanned exploration and assessment - Promotion of international collaboration and technology cooperation by strengthening joint R&D between domestic stakeholders and foreign research institutions